Mathematics Extended Essay on the topic of Erdős' Probabilistic method

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1 Introduction

- A brief introduction - My motivation to explore this topic

2 Scale Free Networks and Graph Theory

- Describing the concept of graphs - Defining terms - Applying the idea to scale free networks

3 The Probabilistic Method

- Describing Probabilistic Notation - Applying using simple a simple Ramsey number example (R(3,3) = 6)

4 Predicting clique sizes in Scale Free Networks

- Setting out the problem: we want to find specific clique sizes in a scale free network with 4 hubs - Using the Chung-Lu Random Graph Model, we will describe how to statitcally generate a graph, with different probabilities of nodes and edges occuring. - Using the first moment we will explain the concept of

5 Building Scale Free Networks

- Using the help of AI to code a python script, I aim to generate a statistically meaningful number of Scale-Free Networks. - By analysing the sizes of clique's present, I should be able to find similar results to my theoretical/probabilitic approach

6 Comparing Results

- Comparing results and making sure theory aligns with empirical data

7 Conclusion

- Conclude on the uses of my results